# Yuxuan Huang

**Phone:** (602)7435746 | **Email:** yhuan591@asu.edu
**LinkedIn:** https://www.linkedin.com/in/yuxuanhuang200204/ | **GitHub:** https://github.com/xander-Huang-ASU

## RESEARCH INTERESTS

LLM agents and multi-agent systems, reliable and controllable agent reasoning, agent evaluation

## EDUCATIONAL BACKGROUND

**Arizona State University (ASU)** **Tempe, USA**
*Master of Science in Computer Science* 08/2025 - Present
- GPA: 3.72/4.0

**Changchun University of Science and Technology (CUST)** **Changchun City, China**
*Bachelor of Engineering in Information Security* 09/2020 - 06/2024
- GPA: 3.54/5.0 (WES equivalent: 3.54/4.0)
- Rank: 6 / 68
- Scholarships: CUST Second-class and Third-class Scholarship (6 consecutive semesters), Academic Competition Scholarship, Lab Project Scholarship, Social Work Scholarship

## RESEARCH EXPERIENCES

**Research Intern, Data Mining and Reinforcement Learning (DaRL) Group, ASU** **05/2026 - Present**
**Advisor:** Hua Wei
**Project:** Evaluating and Improving Reusable Skills for LLM Agents
- Investigated how the design and quality of reusable skills affect the reliability of LLM agent execution, aiming to identify which skill properties make agent behavior more predictable and successful
- Developed a structured evaluation framework capturing execution-relevant skill properties (workflow completeness, validation, inter-step handoffs, error handling, and scope–complexity alignment), and applied controlled perturbations to each property to isolate its effect on agent performance
- Ran repeated agent executions on SkillsBench across diverse tasks and models, using within-task comparisons and statistical association analyses to quantify how changes in skill quality translate into task success across runs

**Research Assistant, Department of Information Systems, ASU** **08/2025 - Present**
**Project 1:** Multimodal Multi-Agent Video Content Moderation
**Advisor:** Che-Wei Liu, Mingwei Sun (University of Texas at Arlington)
- Investigating multimodal content moderation for videos that jointly reasons over vocal tone, transcribed text, and visual keyframes, extending prior work that has largely focused on image–text inputs
- Developing a multi-agent framework in which a dedicated agent analyzes each modality and the agents collaborate to reach a final moderation decision, aiming to improve detection accuracy over single-model and image–text baselines

**Project 2:** Generative AI vs. Marketplace Search: Product Recommendation Study
**Advisor:** Che-Wei Liu
*Pipeline and data supported a paper accepted at CIST 2026, DSI 2026, and WORTH 2027*
- Designed a controlled evaluation comparing product recommendations from generative AI systems (GPT, Grok, Claude, Gemini) against Etsy's native search, sending identical queries in parallel under varying ranking conditions (relevance, price, recency, review volume) with every query paired across channels
- Built fault-tolerant, resumable data infrastructure for large-scale LLM querying and web collection, including graph-based traversal of 4,000+ category nodes and collection of 300,000+ pages
- Validated the extraction pipeline against a hand-labeled gold set before scaling to 51,898 listings, and analyzed the search discoverability of 20,000+ AI-recommended products to characterize what generative AI systems surface

**Additional Contributions:** Online experiment infrastructure for behavioral studies
- Built online experiment platforms supporting multiple behavioral studies, including autonomous-vehicle field and lab experiments (Major Revision, *Information Systems Research*) and AI co-created music evaluation (INFORMS 2026, POMS 2027, WORTH 2027)

**Research Assistant, CUST Network and Information Systems Laboratory**
**Supervisor:** Professor Hui Qi
**Project:** Academic Project Management System 06/2022 - 06/2023
- Developed full-stack features for an academic project management platform, including project submission and scoring modules (Java Spring Boot backend, JavaScript frontend)
- Maintained the backend SQL Server database supporting high-volume academic records
- Gathered system requirements from faculty and academic stakeholders, and iterated on the design based on their feedback to improve usability

- Supported the platform's launch to 100,000+ users during the 2022–2023 academic year, including training faculty members on system usage

## INDUSTRY EXPERIENCE

**Guangdong Sunram Electronic Technology Co., Ltd.** **Foshan City, China**
*Software Development Engineer* 07/2024 - 04/2025

- Trained a YOLOv8 model to recognize 9 types of electronic components on SMT boards with 90%+ accuracy; compiled the training dataset, reducing repetitive labeling work for SOP specialists
- Independently developed an automated error detection program for production-line FCT equipment, using traditional computer vision techniques to analyze camera frames, trigger alerts, and upload screenshots to the MES database for engineers to review
- Independently developed an Android app for material transport (Vue/CSS frontend, Java Spring Boot backend); added scan validation to prevent duplicate entries and redesigned the workflow so that 80% of operations can be completed by QR code scanning
- Wrote SQL scripts to manage company databases (OA, MES, ERP; 150 GB+ on average), extracted and visualized data, and designed frontend pages embedded in OA and BI systems to support management decisions

## PROFESSIONAL SKILLS

**Programming:** Python, C, SQL, Java, JavaScript
**Frameworks & Tools:** PyTorch, Spring Boot, Flask, Vue.js, React
**Databases:** MySQL, SQL Server, Redis
**Languages:** English, Mandarin, Cantonese
**Others:** Alibaba Cloud, AWS Cloud, Linux

## HONORS AND AWARDS

Provincial First Prize, Contemporary Undergraduate Mathematical Contest in Modeling 12/2022
Intramural Second Prize, Contemporary Undergraduate Mathematical Contest in Modeling 08/2022
Second Prize (Northeast Division), WeChat Mini Program Application Development Contest 04/2022

## INTERESTS

- Chinese pastry (certified pastry chef)
- hands-on electrical work (certified electrician)
- video production